GRANT NEILSON

Computational Biologist with seven years of industry experience building scalable and robust analysis pipelines, developing internal Python & R packages and processing large-scale whole genome and single cell sequencing datasets (1TB+). Committed to writing production-quality code through rigorous testing, version control, and thorough documentation. Active open-source contributor, passionate about developing reliable scientific software and advancing reproducible research through community-driven collaboration.

INDUSTRY EXPERIENCE

Senior Computational Biologist

2026 | 2023

CoSyne Therapeutics — London, UK

- Led analysis of large-scale genomic datasets to support biomarker discovery and target id, translating complex results into clear, decision-ready outputs for cross-functional teams. Culminated in the successful partnership with a publicly traded biotech.
- Co-Developed and maintained internal & open source R and Python packages. For statistical modeling, data processing and visualisation. Bixverse, is a Rust based R package enabling the analysis of 3 million single cells on a standard laptop. Cosyne-cnvutils is a python package used for the analysis of genomics data.
- Developed internal container system using Docker and ECR, to enable reproducible research. This system used CI/CD to build images robustly and securely.
- Recognized as a named contributor in version 3.4.1 of the NF-core Sarek pipeline. Optimised the running of the pipeline, achieving a 50% cost reduction and 30% decrease in computational time per sample.

Computational Biologist

2023 | 2020

Fios Genomics — Edinburgh, UK

- Directed projects focusing on high-dimensional omics data types (WES, WGS, scRNA-seq, proteomics), delivering comprehensive analyses and reports for clients.
- Developed in-depth knowledge of statistical methodologies, mastering model selection and fitting techniques, including SVM, elastic net, random forest, and linear/logistic regression, while accounting for covariates.
- Actively contributed to internal R&D by developing and maintaining specialized R packages, enhancing the team's analytical capabilities.
- Led client calls, effectively communicating complex analytical results, ensuring clarity and client understanding.

Computational Biologist

2020 | 2019

Complex Diseases Epigenetics group, Exeter Medical School — Exeter, UK

- Developed novel bioinformatics pipelines for processing genomics data.
- Gained an understanding of regression analysis, how to account for covariates, feature selection and optimal model fitting.

View this CV online

CONTACT

- grant.neilson5@gmail.com
- GitHub
- LinkedIn

LANGUAGES

- R
- Python
- Bash
- Nextflow

TECHNOLOGIES

- GitLab/GitHub
- Docker
- aws
- Azure

OPEN SOURCE & PUBLIC PROJECTS

- bixverse
- mainfoldsR
- cosyne-cnvutils
- Sarek

Made with the R package pagedown.

The source code is available on GitHub.

Last updated March 2026.

EDUCATION

2019 | 2015

MSCi Biology

University of Southampton — Southampton, UK

- **Grade** : First Class Honors
- Winner of the British Society for Immunology Graduate of the year award.
- Recipient of the summer studentship from the Genetics Society (2017).

2015 | 2013

School

Merchiston Castle Shool — Edinburgh, UK

- A Levels: Biology, Chemistry, Maths

SELECTED PUBLICATIONS

2025

TP53 and RB1 are predictive genetic biomarkers for sensitivity to cytarabine in gliomas

BioRxiv

- L. Laaniste, J. Luginbuehl, M. Trinh, R. Watkins, G. Lueg, **G. Neilson**, et al.
- doi: https://doi.org/10.1101/2025.11.20.688938

2025

Multi modal single cell foundation models via dynamic token adaptation

BioRxiv

- W. Zhao, A. Solaguren-Beascoa, **G. Neilson**, et al.
- doi: https://doi.org/10.1101/2025.04.17.649387

2025

Detecting cell level transcriptomic changes of Perturb seq using Contrastive Fine tuning of Single Cell Foundation Models

BioRxiv

- W. Zhao, A. Solaguren-Beascoa, **G. Neilson**, et al.
- doi: https://doi.org/10.1101/2025.04.17.649395